Title:

Characterizing Nonlinear Fluid Flow in 3D Printed Rock Fractures: Effects of Roughness, Aperture, and Intersection AnglesEffects of Roughness, Aperture, and Intersection Angles

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Abstract

Understanding fluid flow behavior at fracture intersections is vital to geothermal energy extraction, underground water flow, and CO_2 storage. Here, laboratory experiments were performed to study the effect of surface roughness, intersection angles, and aperture on nonlinear flow behavior in simple fracture networks. A FormLabs (3B+) 3D printer was used to fabricate 8 samples of 38.1mm in diameter and 59.43mm or 54.9mm in length with 2 intersecting fractures of 10^o and 40^o respectively. The surface roughness of the fractures was based on laser profilometry measurements from two induced mode 1 rock fractures, with a joint roughness coefficient (JRC) of 5 (smoother) and 10 (rougher). The mean apertures of 1mm and 0.7mm for JRC values of 5 and 10, respectively, with an induced mismatch of 1mm. Hydraulic tests were conducted on the intersecting fractures for a range of constant flow rates (20 to 50 ml/min) using a horizontal hydrostatic core holder, and a confining pressure of 1 MPa. The differential pressure was measured to determine the velocity and Reynolds number, Re, to assess the fluid flow behavior through the intersecting fractures.
Re was observed to range from 4.7 to 25.30. An increase in the intersection angle resulted in a decrease in Re, while an increase in roughness results in an increase in Re. The relationship between hydraulic gradient and flow rate is well-described by Forchheimer's law and the Izbash equation, with a high R^2 value of 0.99. The Izbash exponent is observed to increase with the angle of fracture intersection for both JRC values, indicating a deviation from Darcy flow behavior as the fracture intersection angle increases. This suggests that other factors, such as inertial effects, become more influential in governing the flow rate through a fractured medium. Similarly, the Forchheimer coefficient "b" also increases with the angle for both JRC values, indicating that at higher fracture intersections, there is a greater resistance to flow that is attributed to inertial forces. The contribution of viscous forces represented by the Forchheimer coefficient "a" exhibits some variation with respect to the fracture angle and JRC value, lacking a clear and consistent trend. Understanding this behavior is important for the prediction and management of fluid flow through fractured geological formations